

Expenditure Minimization and Hicksian Demand

JEB104 Microeconomics I
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Expenditure Minimization

- consumer's utility function $u(x)$, prices p are given
- consumer wants to reach at least a given utility level ($\bar{u}$)
- seeking a bundle that minimizes expenditures consumer spends to reach the utility $\bar{u}$:

$$\min_{x_1, \dots, x_n} \sum_{i=1}^n p_i x_i$$

$$\text{s.t. (i) } u(x) \geq \bar{u}$$

$$\text{(ii) } x_i \geq 0, i = 1, \dots, n$$

Solution: Hicksian demand function $x_i^* = h_i(p, u)$

Previously

- Standard approach: consumer maximizes utility given his budget constraint (spends entire income if preferences are monotonic)

Today

- A “dual” approach: minimize expenses given some minimum level of utility

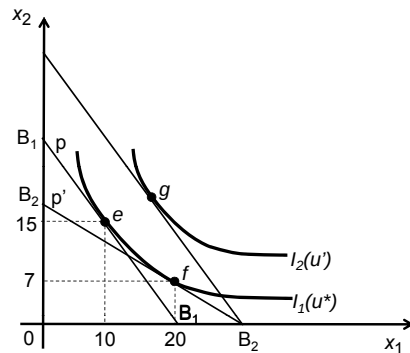
DRAW FIGURE

- Draw several parallel budget lines corresponding to the same relative prices, but different levels of income = isocost curves/lines = isoquants of expenditures – each line represents a given level of expenditures
- One indifference curve corresponding to $\bar{u}$ – consumer wants to reach at least this level of utility
- Which bundle minimizes costs on reaching $\bar{u}$? The bundle where the indifference curve touches the lowest isoquant of expenditures.

 $h_i(p, u)$

- A function of prices and utility (as opposed to prices and income)
- For each price vector and utility level, the Hicksian demand assigns a bundle that allows a consumer to reach this particular utility level with the least costs.

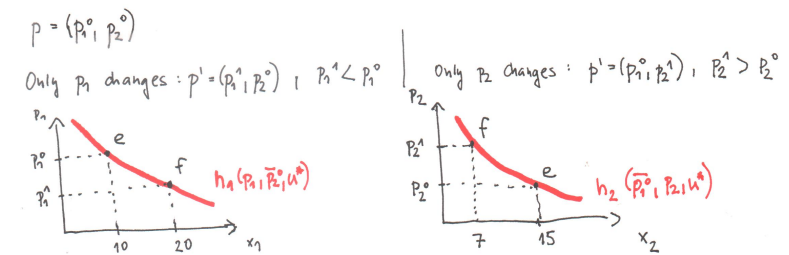
Expenditure Minimization



Optimum:

- e to reach u^* when prices are p
- f to reach u^* when prices are p'
- g to reach u' when prices are p

Change from p to p' can be due to change in p_1 or p_2 :



Hicksian demands and decomposition into IE and SE discussed in the previous lecture

- In this case, p_2 would be going down (decompose the change in demand for x_2 due to change in price of x_2 into SE and IE)
- the original optimum f , new optimum g (both are points on Walrasian demand $D(\bar{p}_1, p_2, m)$)
- $SE = e - f = 15 - 7 = 8$
- $IE = g - e = (\text{could be e.g. } 18) - 15 = 3$
- Movement due to SE is actually movement along Hicksian demand (utility remains constant)! (f and e are points on Hicksian demand $h(\bar{p}_1, p_2, u^*)$)

Expenditure Minimization: Interior Optimum

Lagrangian function:

$$\mathcal{L}(x_1, x_2, \dots, x_n, \lambda) = \sum_{i=1}^n p_i x_i + \lambda(\bar{u} - u(x_1, x_2, \dots, x_n))$$

First-order conditions:

$$\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial x_i} = p_i - \lambda^* \frac{\partial u(\mathbf{x}^*)}{\partial x_i} = 0, \quad \forall i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}(\mathbf{x}^*, \lambda^*)}{\partial \lambda} = \bar{u} - u(\mathbf{x}^*) = 0$$

Rewriting FOCs:

$$\frac{\lambda^* \partial u(\mathbf{x}^*)}{\partial x_i} = p_i$$

For any $i, j \in 1, \dots, n$, $i \neq j$:

$$\frac{\frac{\partial u(\mathbf{x}^*)}{\partial x_i}}{\frac{\partial u(\mathbf{x}^*)}{\partial x_j}} = \frac{MU_i(\mathbf{x}^*)}{MU_j(\mathbf{x}^*)} = \frac{\lambda^* p_i}{\lambda^* p_j} = \frac{p_i}{p_j} \quad \equiv \quad \text{Optimality condition}$$

Very similar to utility maximization, only “swapped”

Lagrange multiplier λ

- p_i / MU_i - cost for increasing utility by one util
- the minimum amount we would have to pay to increase our utility by one util

In interior optimum (consuming positive amounts of all the goods), the optimality condition well-known from the utility maximization problem must hold, too (optimum is a point, in which an indifference curve representing the utility level $\bar{u}$ touches the lowest isocost line – line representing a particular level of spending with slope $-\frac{p_1}{p_2}$).

Expenditure Minimization: Example

Multiplicative utility function $u(x_1, x_2) = x_1 x_2$

Optimization problem:

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2$$

$$\text{s.t. } x_1 x_2 \geq \bar{u}$$

Lagrangian:

$$\mathcal{L}(x_1, x_2, \lambda) = p_1 x_1 + p_2 x_2 + \lambda(\bar{u} - x_1 x_2)$$

Expenditure Minimization: Example

$$\mathcal{L}(x_1, x_2, \lambda) = p_1 x_1 + p_2 x_2 + \lambda(\bar{u} - x_1 x_2)$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial x_1} = p_1 - \lambda x_2 = 0 \quad \Rightarrow \quad p_1 = \lambda x_2$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = p_2 - \lambda x_1 = 0 \quad \Rightarrow \quad p_2 = \lambda x_1$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \bar{u} - x_1 x_2 = 0 \quad \Rightarrow \quad \bar{u} = x_1 x_2$$

Hence, demand functions are

$$h_1(p_1, p_2, u) = \sqrt{\frac{p_2 u}{p_1}} \quad h_2 = \sqrt{\frac{p_1 u}{p_2}}$$

Calculation:

Divide the first two FOCs: $\frac{p_1}{p_2} = \frac{x_2}{x_1}$

$$x_2 = x_1 \cdot \frac{p_1}{p_2} \Rightarrow \bar{u} = x_1^2 \cdot \frac{p_1}{p_2}$$

Hence,

$$x_1^* = \sqrt{\frac{p_2 \bar{u}}{p_1}} \quad x_2^* = \sqrt{\frac{p_1 \bar{u}}{p_2}}$$

In practice we ignore positivity constraints in the problem

- So if we get $x_i^* < 0 \Rightarrow x_i^* = 0$ and we get a corner solution and have to recalculate x_j^*

Expenditure Minimization: Corner Solution

$$x_j^* = 0 : \quad MRS_{ji}(x^*) = -\frac{MU_i(x^*)}{MU_j(x^*)} \leq -\frac{p_i}{p_j}$$

$$\Rightarrow \frac{p_i}{MU_i(x^*)} \leq \frac{p_j}{MU_j(x^*)}$$

$$\mathcal{L}(x_1, x_2, \dots, x_n, \lambda) = \sum_{i=1}^n p_i x_i + \lambda(\bar{u} - u(x_1, x_2, \dots, x_n))$$

Kuhn-Tucker (complementary slackness) conditions:

For all $i = 1, \dots, n$,

$$1. \quad \frac{\partial \mathcal{L}(x^*, \lambda^*)}{\partial x_i} = p_i - \lambda^* \frac{\partial u(x^*)}{\partial x_i} \geq 0 \quad \wedge \quad x_i^* \geq 0 \quad \wedge \quad x_i^* \cdot \left(p_i - \lambda^* \frac{\partial u(x^*)}{\partial x_i} \right) = 0;$$

$$2. \quad \frac{\partial \mathcal{L}(x^*, \lambda^*)}{\partial \lambda} = \bar{u} - u(x^*) \leq 0 \quad \wedge \quad \lambda^* \geq 0 \quad \wedge \quad \lambda^* \cdot (\bar{u} - u(x^*)) = 0.$$

Hicksian Demand $h(p, u)$

Also called Compensated demand (price change compensated by income in order to keep the same utility)

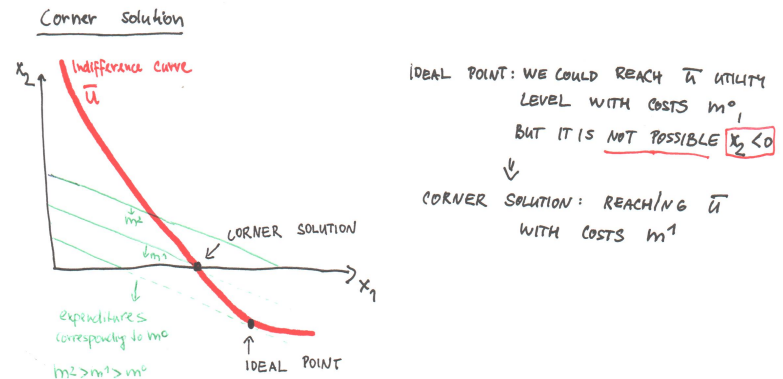
- 1 Non-increasing in own price: $\frac{\partial h_i(p, u)}{\partial p_i} \leq 0$
- 2 Homogeneous of degree zero in prices p : $h(\alpha p, u) = h(p, u)$, $\alpha > 0$
- 3 No excess utility: for any $x \in h(p, \bar{u})$ $u(x) = \bar{u}$
- 4 Uniqueness: If $u(x)$ is strictly quasiconcave, then $h(p, u)$ consists of single element (unique $h(p, u)$ for any combination (p, u))
- 5 Cross-substitution effects are symmetric: $\frac{\partial h_i(p, u)}{\partial p_j} = \frac{\partial h_j(p, u)}{\partial p_i}$

Corner solution

- Cost for increasing utility by one util is higher for good j than for good i
- Hence it is beneficial to move consumption to good i instead of good j , he can get the same utility level with lower cost.
- equality in the first equation illustrates the case when optimality condition holds for corner solution, i.e. tangency point is exactly the point where $x_j = 0$

Explanation of inequality signs that it cannot be the other way around:

1. if $p_i < \lambda^* \frac{\partial u(x^*)}{\partial x_i}$ consumer could get more utility for a lower price
2. if $\bar{u} > u(x^*)$ then the constraint is broken



Non-increasing

- Consumer never increases demand for good, whose price increases to keep the same utility when minimizing spending
- Follows from convexity of preferences
- Note that it is exactly equivalent to the statement that substitution effect is always negative!

Homogeneous of degree zero in prices

- when all prices are multiplied by the same coefficient, relative prices do not change, optimal bundle does not change
- what only changes is the costs of reaching the optimum

Symmetric

- $\frac{\partial h_i(p, u)}{\partial p_j}$ is exactly cross-substitution effect (remember we can study change in demand for x_i due to change in p_j and decompose this change to cross-substitution effect (CSE) and income effect (see the previous lecture)
- Goods are either net complements $\frac{\partial h_i(p, u)}{\partial p_j} < 0$ or net substitutes $\frac{\partial h_i(p, u)}{\partial p_j} > 0$
- It cannot be that one is net complement of other and the other is net substitute of the first one

You can check by yourselves that the Hicksian demands computed above satisfy all these conditions.

Example: Hicksian Demand Used to Derive SE and IE

Suppose that consumer has utility function $u(x_1, x_2) = x_1 x_2$:

- ➊ Derive general Marshallian and Hicksian demands: $x_1(p_1, p_2, m)$, $x_2(p_1, p_2, m)$, $h_1(p_1, p_2, u)$, $h_2(p_1, p_2, u)$
- ➋ Suppose that $p_1^0 = 8$, $p_2 = 2$, $m = 80$, what are the quantities of x_1 and x_2 demanded?
- ➌ Suppose that the price of x_1 changes to $p_1^1 = 2$, what are the quantities of x_1 and x_2 demanded?
- ➍ Decompose the total effect upon the demand for x_1 into substitution and income effect.
- ➎ Back to the original case, now suppose that the price of x_1 changes to $p_1^2 = 16$, what are the quantities of x_1 and x_2 demanded?
- ➏ Decompose the total effect upon the demand for x_1 into substitution and income effect.

1. $h_1(p_1, p_2, u) = \sqrt{\frac{p_2 u}{p_1}}$, $h_2 = \sqrt{\frac{p_1 u}{p_2}}$, $x_1(p_1, p_2, m) = \frac{m}{2p_1}$, $x_2(p_1, p_2, m) = \frac{m}{2p_2}$
2. $x_1^0 = 5$, $x_2^0 = 20$
3. $x_1^1 = 20$, $x_2^1 = x_2^0 = x_2 = 20$
4. Total effect (change in demand for x_1) $TE = 20 - 5 = 15$. To decompose TE to SE and IE, we have to compute $h_1(p_1, p_2, u)$ for new price combination, i.e. $p_1^1 = 2$, $p_2 = 2$ and original utility level.
 - Original level of utility in optimum (x_1^0, x_2^0) is $u(x_1, x_2) = 5 \cdot 20 = 100$
 - $h_1(2, 2, 100) = \sqrt{\frac{2 \cdot 100}{2}} = 10$
 - Hence, substitution effect is $SE = 10 - 5 = 5$ and income effect is $IE = 20 - 10 = 10$.
5. $x_2^1 = 2.5$, $x_2^2 = x_2^0 = x_2 = 20$
6. Total effect (change in demand for x_1) $TE = 2.5 - 5 = -2.5$. To decompose TE to SE and IE, we have to compute $h_1(p_1, p_2, u)$ for new price combination, i.e. $p_1^2 = 16$, $p_2 = 2$ and original utility level.
 - Original level of utility in optimum (x_1^0, x_2^0) is $u(x_1, x_2) = 5 \cdot 20 = 100$
 - $h_1(16, 2, 100) = \sqrt{\frac{2 \cdot 100}{16}} = \frac{5\sqrt{2}}{2} \simeq 3.54$
 - Hence, substitution effect is $SE = \frac{5\sqrt{2}}{2} - 5 \simeq -1.46$ and income effect is $IE = 2.5 - \frac{5\sqrt{2}}{2} \simeq -1.04$.